

Class 12 Equivalence Relations

Goals:

1. Understand connection between partition of a set, S , and an equivalence relation on S .
2. Be able to prove something is an equivalence relation.

Definition: If a is any integer and n is any positive integer we define the congruence class containing $a \bmod n$ to be

Example: List all the elements in the congruence class $[0]_5$.

$$[0]_5 =$$

How many distinct congruence classes are there mod n ?

Finish the sentence: $[a]_n = [b]_n$ if and only if

Crucial idea: The congruence classes mod n partition the set of integers into n subsets.

We can think of the set $\mathbf{Z}_n$ as being the set of all distinct congruence classes mod n and define modular addition and multiplication as adding and multiplying these classes.

Equivalence Relations

Let S be a nonempty set and let P be any partition of S , into nonempty subsets $S_1, S_2, \dots, S_n$. This partition will define an equivalence relation on S in the following way.

The subsets are called the equivalence classes for the relation. The equivalence classes will always form a partition of the set S .

Alternate definition: An equivalence relation on a set S is a nonempty set of ordered pairs in $S \times S$ that satisfies three properties for all a , b , and c in S .

$\sim$ satisfies the reflexive property

$\sim$ satisfies the symmetric property

$\sim$ satisfies the transitive property

Example: Let S be the set of students in this class. Define a relation on this set as follows: $a \sim b$ if and only if a is born in the same month as b .

Reflexive

Symmetric

Transitive

Describe the equivalence classes.

Practice: Problem #60 on page 116

Let S be the set of all integers. For any a and b in $\mathbf{Z}$ define $a \sim b$ if and only if $a + b$ is even. Prove $\sim$ is an equivalence relation on $\mathbf{Z}$ and describe the equivalence classes.

Create: Choose any set S and define your own relation on it.